

I, J, A 是任意集合, 则有

$$A^{I \times J} = \{\text{映射 } f: I \times J \rightarrow A\}$$

$$A^I = \{\text{映射 } f: I \rightarrow A\} \quad A^J = \{\text{映射 } f: J \rightarrow A\}$$

$$(A^I)^J = \{\text{映射 } f: J \rightarrow A^I\} \quad (A^J)^I = \{\text{映射 } f: I \rightarrow A^J\}$$

问题 1: 给出自然的双射 $A^{I \times J} \xrightarrow{1:1} (A^I)^J$, 其中 I, J, A 是任意集合

解: 定义映射 $\varphi: A^{I \times J} \longrightarrow (A^I)^J$

$$f \longmapsto \varphi(f): J \rightarrow A^I$$

$$j \longmapsto \varphi(f)(j): I \rightarrow A$$

$$i \longmapsto f(i, j)$$

对 $\forall f \in A^{I \times J}$, 有: f 是 $I \times J \rightarrow A$ 的映射.

我们需要证明 $\varphi(f): J \rightarrow A^I$

确实是 $J \rightarrow A^I$ 的映射

$$j \longmapsto \varphi(f)(j): I \rightarrow A$$

$$i \longmapsto f(i, j)$$

对 $\forall j \in J$, $\varphi(f)(j): I \rightarrow A$
 $i \longmapsto f(i, j).$

对 $\forall i \in I$, $\because i \in I$ 且 $j \in J \therefore f(i, j) \in A \therefore \varphi(f)(j)(i) \in A$

对 $\forall i_1, i_2 \in I$, 若 $i_1 = i_2$, 则 $(i_1, j) = (i_2, j) \therefore f(i_1, j) = f(i_2, j)$

$$\varphi(f)(j)(i_1) = f(i_1, j) = f(i_2, j) = \varphi(f)(j)(i_2)$$

$\therefore \varphi(f)(j)$ 是 $I \rightarrow A$ 的映射 $\therefore \varphi(f)(j) \in A^I$

对 $\forall j_1, j_2 \in J$, 若 $j_1 = j_2$, 则有: ~~$\varphi(f)(j_1) = \varphi(f)(j_2)$~~

$$\varphi(f)(j_1): I \rightarrow A$$
$$v \mapsto f(v, j_1)$$

$$\varphi(f)(j_2): I \rightarrow A$$
$$v \mapsto f(v, j_2)$$

$$\because j_1 = j_2 \quad \therefore (v, j_1) = (v, j_2) \quad \therefore f(v, j_1) = f(v, j_2)$$

$$\text{对 } \forall v \in I, \quad \varphi(f)(j_1)(v) = f(v, j_1) = f(v, j_2) = \varphi(f)(j_2)(v)$$

$$\therefore \varphi(f)(j_1) = \varphi(f)(j_2)$$

$$\therefore \varphi(f) \text{ 是 } J \rightarrow A^I \text{ 的映射} \quad \therefore \varphi(f) \in (A^I)^J$$

~~对 $\forall j_1, j_2 \in J$, 若 $j_1 = j_2$, 则有:~~

对 $\forall f_1, f_2 \in A^{I \times J}$, 若 $f_1 = f_2$, 则有:

$$\varphi(f_1): J \rightarrow A^I$$
$$j \mapsto \varphi(f_1)(j): I \rightarrow A$$
$$v \mapsto f_1(v, j)$$

$$\varphi(f_2): J \rightarrow A^I$$
$$j \mapsto \varphi(f_2)(j): I \rightarrow A$$
$$v \mapsto f_2(v, j)$$

$$\text{对 } \forall j \in J, \quad \varphi(f_1)(j): I \rightarrow A \quad \varphi(f_2)(j): I \rightarrow A$$
$$v \mapsto f_1(v, j) \quad v \mapsto f_2(v, j)$$

$$\text{对 } \forall v \in I, \quad \varphi(f_1)(j)(v) = f_1(v, j) = f_2(v, j) = \varphi(f_2)(j)(v)$$

$$\therefore \varphi(f_1)(j) = \varphi(f_2)(j) \quad \therefore \varphi(f_1) = \varphi(f_2)$$

$\therefore \varphi$ 是 $A^{I \times J} \rightarrow (A^I)^J$ 的映射.

对 $\forall f_1, f_2 \in A^{I \times J}$, 若 $\varphi(f_1) = \varphi(f_2)$, 则有:

$$\varphi(f_1): J \rightarrow A^I$$
$$j \mapsto \varphi(f_1)(j): I \rightarrow A$$
$$v \mapsto f_1(v, j)$$

$$\varphi(f_2): J \rightarrow A^I$$
$$j \mapsto \varphi(f_2)(j): I \rightarrow A$$
$$v \mapsto f_2(v, j)$$

对 $\forall (i, j) \in I \times J$, 有: $i \in I$ 且 $j \in J$

$$\because \varphi(f_1) \in (A^I)^J \quad \therefore \varphi(f_1)(j) \in A^I$$

$$\because \varphi(f_2) \in (A^I)^J \quad \therefore \varphi(f_2)(j) \in A^I$$

$$\because \varphi(f_1) = \varphi(f_2) \quad \therefore \varphi(f_1)(j) = \varphi(f_2)(j)$$

$$\because i \in I \quad \therefore \varphi(f_1)(j)(i) = \varphi(f_2)(j)(i)$$

$$\therefore f_1(i, j) = f_2(i, j) \quad \therefore f_1 = f_2$$

$\therefore \varphi$ 是 $A^{I \times J} \rightarrow (A^I)^J$ 的单射.

对 $\forall h \in (A^I)^J$, 有: h 是 $J \rightarrow A^I$ 的映射.

定义映射 $f: I \times J \rightarrow A$
 $(i, j) \mapsto h(j)(i).$

对 $\forall (i, j) \in I \times J$, ~~有~~ 有: $i \in I$ 且 $j \in J$

$$\because j \in J \quad \therefore h(j) \in A^I \quad \because i \in I \quad \therefore h(j)(i) \in A$$

$$\therefore f(i, j) = h(j)(i) \in A$$

对 $\forall (i_1, j_1), (i_2, j_2) \in I \times J$, 若 $(i_1, j_1) = (i_2, j_2)$, 则

~~有~~ $i_1 = i_2$ 且 $j_1 = j_2$ $\because h$ 是 $J \rightarrow A^I$ 的映射, $j_1 = j_2 \in J$

$$\therefore h(j_1) = h(j_2) \quad \therefore h(j_1) = h(j_2) \in A^I$$

$$\therefore h(j_1)(i_1) = h(j_1)(i_2) = h(j_2)(i_2)$$

$\therefore f(i_1, j_1) = f(i_2, j_2) \quad \therefore f$ 是 $I \times J \rightarrow A$ 的映射. $\therefore f \in A^{I \times J}$

$$\therefore \varphi(f) : J \rightarrow A^I \quad (\text{或者写: } \therefore f \in A^{I \times J} \therefore \varphi(f) \in (A^I)^J \text{ 且 } h \in (A^I)^J)$$

$$j \mapsto \varphi(f)(j) : I \rightarrow A$$

$$i \mapsto f(i, j)$$

$$\text{对 } \forall j \in J, \quad \varphi(f)(j) : I \rightarrow A \quad h(j) \in A^I$$

$$i \mapsto f(i, j)$$

$$\text{对 } \forall i \in I, \quad \varphi(f)(j)(i) = f(i, j) = h(j)(i)$$

$$\therefore \varphi(f)(j) = h(j) \quad \therefore \varphi(f) = h$$

$$\therefore \varphi \text{ 是 } A^{I \times J} \rightarrow (A^I)^J \text{ 的满射}$$

$$\therefore \varphi \text{ 是 } A^{I \times J} \rightarrow (A^I)^J \text{ 的双射.}$$

至于 φ 为什么是“自然的”双射，之后学习范畴论相关知识，再补全。 $\square$

问题 2: 给出自然的双射 $A^{I \times J} \xrightarrow{1:1} (A^J)^I$, 其中 I, J, A 是任意集合

$$\text{解: 定义映射 } \varphi : A^{I \times J} \longrightarrow (A^J)^I$$

$$f \mapsto \varphi(f) : I \longrightarrow A^J$$

$$i \mapsto \varphi(f)(i) : J \longrightarrow A$$

$$j \mapsto f(i, j)$$

$$\text{对 } \forall f \in A^{I \times J}, \text{ 有: } f \text{ 是 } I \times J \rightarrow A \text{ 的映射}$$

$$\text{对 } \forall i \in I, \quad \varphi(f)(i) : J \rightarrow A$$

$$j \mapsto f(i, j)$$

$$\text{对 } \forall j \in J, \quad \varphi(f)(i)(j) = f(i, j) \quad \because i \in I \text{ 且 } j \in J \quad \therefore (i, j) \in I \times J$$

$$\therefore f(i, j) \in A \quad \therefore \varphi(f)(i)(j) = f(i, j) \in A$$

对 $\forall j_1, j_2 \in J$, 若 $j_1 = j_2$, 则 $(i, j_1) = (i, j_2) \therefore f(i, j_1) = f(i, j_2)$

$$\therefore \varphi(f)(i)(j_1) = f(i, j_1) = f(i, j_2) = \varphi(f)(i)(j_2)$$

$$\therefore \varphi(f)(i) \in A^J$$

对 $\forall i_1, i_2 \in I$, 若 $i_1 = i_2$, 则 $\therefore i_1 \in I \therefore \varphi(f)(i_1) \in A^J$

$$\therefore i_2 \in I \therefore \varphi(f)(i_2) \in A^J$$

对 $\forall j \in J$, $\therefore i_1 = i_2 \therefore (i_1, j) = (i_2, j) \therefore f(i_1, j) = f(i_2, j)$

$$\therefore \varphi(f)(i_1)(j) = f(i_1, j) = f(i_2, j) = \varphi(f)(i_2)(j)$$

$$\therefore \varphi(f)(i_1) = \varphi(f)(i_2)$$

$$\therefore \varphi(f) \in (A^J)^I$$

对 $\forall f_1, f_2 \in A^{I \times J}$, 若 $f_1 = f_2$, 则 $\therefore f_1 \in A^{I \times J} \therefore \varphi(f_1) \in (A^J)^I$

$$\therefore f_2 \in A^{I \times J} \therefore \varphi(f_2) \in (A^J)^I$$

对 $\forall i \in I$, $\therefore \varphi(f_1) \in (A^J)^I \therefore \varphi(f_1)(i) \in A^J$

$$\therefore \varphi(f_2) \in (A^J)^I \therefore \varphi(f_2)(i) \in A^J$$

$$\text{对 } \forall j \in J, \varphi(f_1)(i)(j) = f_1(i, j) = f_2(i, j) = \varphi(f_2)(i)(j)$$

$$\varphi(f_1)(i) = \varphi(f_2)(i) \therefore \varphi(f_1) = \varphi(f_2)$$

$\therefore \varphi$ 是 $A^{I \times J} \rightarrow (A^J)^I$ 的映射.

对 $\forall f_1, f_2 \in A^{I \times J}$, 若 $\varphi(f_1) = \varphi(f_2)$, 则: 对 $\forall (i, j) \in I \times J$

$$\text{有: } i \in I \text{ 且 } j \in J \therefore f_1 \in A^{I \times J} \therefore \varphi(f_1) \in (A^J)^I \therefore \varphi(f_1)(i) \in A^J$$

$$\therefore f_2 \in A^{I \times J} \therefore \varphi(f_2) \in (A^J)^I \therefore \varphi(f_2)(i) \in A^J$$

$$\because \varphi(f_1) = \varphi(f_2) \quad \therefore \varphi(f_1)(i) = \varphi(f_2)(i) \in A^J \quad \because j \in J$$

$$\therefore \varphi(f_1)(i)(j) = \varphi(f_2)(i)(j) \quad \therefore f_1(i, j) = f_2(i, j)$$

$$\therefore f_1 = f_2 \quad \therefore \varphi \text{ 是 } A^{I \times J} \rightarrow (A^J)^I \text{ 的单射.}$$

对 $\forall h \in (A^J)^I$, 有: h 是 $I \rightarrow A^J$ 的映射.

$$\text{定义映射 } f: I \times J \rightarrow A \\ (i, j) \mapsto h(i)(j)$$

$$\text{对 } \forall (i, j) \in I \times J, \text{ 有: } i \in I \text{ 且 } j \in J \quad \because i \in I \quad \therefore h(i) \in A^J$$

$$\because j \in J \quad \therefore h(i)(j) \in A \quad \therefore f(i, j) = h(i)(j) \in A$$

对 $\forall (i_1, j_1), (i_2, j_2) \in I \times J$, 若 $(i_1, j_1) = (i_2, j_2)$, 则有:

$$i_1 = i_2 \in I, j_1 = j_2 \in J \quad \because h \text{ 是 } I \rightarrow A^J \text{ 的映射, } i_1 = i_2 \in I$$

$$\therefore h(i_1) = h(i_2) \in A^J \quad \therefore h(i_1)(j_1) = h(i_1)(j_2) = h(i_2)(j_2)$$

$$\therefore f(i_1, j_1) = f(i_2, j_2) \quad \therefore f \text{ 是 } I \times J \rightarrow A \text{ 的映射}$$

$$\therefore f \in A^{I \times J} \quad \therefore \varphi(f) \in (A^J)^I$$

$$\text{对 } \forall i \in I, \quad \because h \in (A^J)^I \quad \therefore h(i) \in A^J \quad \therefore \varphi(f) \in (A^J)^I$$

$$\therefore \varphi(f)(i) \in A^J \quad \text{对 } \forall j \in J, \varphi(f)(i)(j) = f(i, j) = h(i)(j)$$

$$\therefore \varphi(f)(i) = h(i) \quad \therefore \varphi(f) = h$$

$$\therefore \varphi \text{ 是 } A^{I \times J} \rightarrow (A^J)^I \text{ 的满射} \quad \therefore \varphi \text{ 是 } A^{I \times J} \rightarrow (A^J)^I \text{ 的双射.}$$

至于 φ 为什么是“自然的”双射, 之后学习范畴论相关知识, 再补全.



~~Lemma~~ Lemma (群作用的存在性) G 是一个任意的群, X 是一个任意的非空集合, 则一定至少存在一个群 G 在非空集合 X 上的作用

Proof: 定义映射 $\alpha: G \times X \rightarrow X$

$$(g, x) \mapsto x$$

则有: 对 $\forall g_1, g_2 \in G, \forall x \in X$, 有: $\because g_1, g_2 \in G \therefore g_1 g_2 \in G$

$$\therefore \alpha(g_1 g_2, x) = x$$

$$\alpha(g_1, \alpha(g_2, x)) = \alpha(g_1, x) = x$$

$$\therefore \alpha(g_1 g_2, x) = \alpha(g_1, \alpha(g_2, x))$$

对 $\forall x \in X$, 有: $\alpha(1_G, x) = x$

$\therefore$ 映射 $\alpha: G \times X \rightarrow X$ 是一个群作用. $\square$

$$(g, x) \mapsto x$$

Lemma: G 和 G' 是任意两个群, 则一定至少存在一个从 G 到 G' 的群同态.

Proof: 定义映射 $f: G \rightarrow G'$, 显然 f 是 $G \rightarrow G'$ 的映射.
$$x \mapsto 1_{G'}$$

$$\text{对 } \forall x, y \in G, \text{ 有: } \overset{xy \in G}{f(xy)} = 1_{G'} = 1_{G'} \cdot 1_{G'} = f(x) \cdot f(y)$$

$\therefore f: G \rightarrow G'$ 是群同态. $\square$
$$x \mapsto 1_{G'}$$

以下先规范一些符号的含义.

G 是一个任意的群, $1_G \in G$ 是群 G 的么元 (因此 $G \neq \emptyset$)

X 是一个任意的非空集合, 则有:

$$X^{G \times X} = \{\text{映射 } f: G \times X \rightarrow X\}, \quad X^X = \{\text{映射 } f: X \rightarrow X\}$$

$$(X^X)^G = \{\text{映射 } f: G \rightarrow X^X\}, \quad \mathcal{S}_X = \{\text{双射 } \sigma: X \rightarrow X\}$$

显然有 $\mathcal{S}_X \subseteq X^X$, 已经证明了 $(\mathcal{S}_X, \circ)$ 是群 (即: $\mathcal{S}_X$ 对映射的合成构成群)

定义集合 $\text{Act}(G, X) = \{a \in X^{G \times X} \mid a \text{ 是群 } G \text{ 在非空集合 } X \text{ 上的左作用}\}$

$$\therefore \text{Act}(G, X) = \{a \in X^{G \times X} \mid (\text{对 } \forall x \in X, \text{ 都有 } a(1_G, x) = x) \text{ 且 } (\text{对 } \forall g_1, g_2 \in G, \forall x \in X, \text{ 都有 } a(g_1 g_2, x) = a(g_1, a(g_2, x)))\}$$

$$\therefore \text{Act}(G, X) \subseteq X^{G \times X}$$

定义集合 $\text{Hom}(G, \mathcal{S}_X) = \{A \in (\mathcal{S}_X)^G \mid A \text{ 是 } G \rightarrow \mathcal{S}_X \text{ 的群同态}\}$

$$\therefore \text{Hom}(G, \mathcal{S}_X) = \{A \in (\mathcal{S}_X)^G \mid \text{对 } \forall g_1, g_2 \in G, \text{ 都有 } A(g_1 g_2) = A(g_1) \circ A(g_2)\}$$

$$\therefore \text{Hom}(G, \mathcal{S}_X) \subseteq (\mathcal{S}_X)^G$$

Recall Lemma: X 和 Y 是任意的两个集合, $f: X \rightarrow Y$ 是一个单射,

$A \subseteq X, B \subseteq Y$, 定义映射 $g: A \rightarrow B$, 则有:
 $x \mapsto f(x)$.

映射 $g: A \rightarrow B$ 良定且是双射 $\Leftrightarrow f(A) = \{f(x) \mid x \in A\} = B$.

$\therefore X^{G \times X}$ 和 $(X^X)^G$ 是 ~~两个~~ 两个集合

$\therefore$ 问题 2 中已证存在自然的双射 $\varphi: X^{G \times X} \xrightarrow{1:1} (X^X)^G$ 为:

$$\varphi: X^{G \times X} \xrightarrow{1:1} (X^X)^G$$

$$a \longmapsto \varphi(a): G \rightarrow X^X$$

$$g \mapsto \varphi(a)(g): X \rightarrow X$$

$$x \mapsto a(g, x)$$

已知 $\text{Act}(G, X) \subseteq X^{G \times X}$

对 $\forall f \in \text{Hom}(G, \mathbb{G}_X)$, 有: $f \in (\mathbb{G}_X)^G$ 且 f 是 $G \rightarrow \mathbb{G}_X$ 的群同态

$\therefore f$ 是 $G \rightarrow \mathbb{G}_X$ 的映射. $\therefore f$ 是 $G \rightarrow X^X$ 的映射.

$\therefore f \in (X^X)^G \quad \therefore \text{Hom}(G, \mathbb{G}_X) \subseteq (X^X)^G$

任取 $\varphi(\text{Act}(G, X))$ 中的一个元素: $\varphi(a)$ (其中 $a \in \text{Act}(G, X)$)

$\therefore a \in \text{Act}(G, X) \subseteq X^{G \times X} \quad \therefore a \in X^{G \times X} \quad \therefore \varphi(a) \in (X^X)^G$

$\therefore \varphi(a)$ 是 $G \rightarrow X^X$ 的映射.

对 $\forall g \in G, \varphi(a)(g) \in X^X \quad \therefore \varphi(a)(g)$ 是 $X \rightarrow X$ 的映射.

对 $\forall x_1, x_2 \in X$, 若 $\varphi(a)(g)(x_1) = \varphi(a)(g)(x_2)$, 则

$a(g, x_1) = a(g, x_2) \in X. \quad \therefore g \in G \quad \therefore g^{-1} \in G.$

$\therefore a(g, x_1) = a(g, x_2) \in X \quad \therefore (g^{-1}, a(g, x_1)) = (g^{-1}, a(g, x_2)) \in G \times X$

$\therefore a$ 是 $G \times X \rightarrow X$ 的映射 $\therefore a(g^{-1}, a(g, x_1)) = a(g^{-1}, a(g, x_2))$

$\therefore a \in \text{Act}(G, X) \quad \therefore a$ 是群 G 在非空集合 X 上的群作用

$\therefore a(g^{-1}g, x_1) = a(g^{-1}g, x_2) \quad \therefore a(1_G, x_1) = a(1_G, x_2) \quad \therefore x_1 = x_2$

$\therefore \varphi(a)(g)$ 是 $X \rightarrow X$ 的单射.

对 $\forall y \in X$, $\therefore g \in G \quad \therefore g^{-1} \in G \quad \therefore (g^{-1}, y) \in G \times X$

$\therefore a \in X^{G \times X} \quad \therefore a$ 是 $G \times X \rightarrow X$ 的映射 $\therefore a(g^{-1}, y) \in X$.

$$\therefore \varphi(a)(g)(a(g^{-1}, y)) = a(g, a(g^{-1}, y)) = a(gg^{-1}, y)$$

$$= a(1_G, y) = y \quad \therefore \varphi(a)(g) \text{ 是 } X \rightarrow X \text{ 的满射}$$

$\therefore \varphi(a)(g)$ 是 $X \rightarrow X$ 的双射. $\therefore \varphi(a)(g) \in \mathcal{S}_X \subseteq X^X$

$\therefore$ 对 $\forall g \in G$, $\varphi(a)(g) \in X^X$ 且有 $\varphi(a)(g) \in \mathcal{S}_X \subseteq X^X$

$\therefore \varphi(a)$ 是 $G \rightarrow \mathcal{S}_X$ 的映射 $\therefore \varphi(a) \in (\mathcal{S}_X)^G$

对 $\forall g_1, g_2 \in G$, 有: $g_1 g_2 \in G$. $\therefore \varphi(a)(g_1 g_2) \in \mathcal{S}_X$,

$$\varphi(a)(g_1) \in \mathcal{S}_X, \varphi(a)(g_2) \in \mathcal{S}_X \quad \therefore \varphi(a)(g_1) \circ \varphi(a)(g_2) \in \mathcal{S}_X.$$

$$\text{对 } \forall x \in X, \varphi(a)(g_1 g_2)(x) = a(g_1 g_2, x) = a(g_1, a(g_2, x))$$

$$(\varphi(a)(g_1) \circ \varphi(a)(g_2))(x) = \varphi(a)(g_1)(\varphi(a)(g_2)(x))$$

$$= \varphi(a)(g_1)(a(g_2, x)) = a(g_1, a(g_2, x))$$

$$\therefore \varphi(a)(g_1 g_2)(x) = (\varphi(a)(g_1) \circ \varphi(a)(g_2))(x)$$

$$\therefore \varphi(a)(g_1 g_2) = \varphi(a)(g_1) \circ \varphi(a)(g_2)$$

$$\therefore \varphi(a) \in \text{Hom}(G, \mathcal{S}_X) \quad \therefore \varphi(\text{Act}(G, X)) \subseteq \text{Hom}(G, \mathcal{S}_X)$$

对 $\forall A \in \text{Hom}(G, \mathcal{S}_X)$, 有: $A \in (\mathcal{S}_X)^G$ 且 A 是 $G \rightarrow \mathcal{S}_X$ 的群同态.

$\therefore A \in (\mathcal{S}_X)^G \quad \therefore A$ 是 $G \rightarrow \mathcal{S}_X$ 的映射 $\therefore A$ 是 $G \rightarrow X^X$ 的映射

$\therefore A \in (X^X)^G \quad \therefore \varphi: X^{G \times X} \xrightarrow{1:1} (X^X)^G$ 是双射 $\therefore$ 存在唯一的 $a \in X^{G \times X}$, s.t. $\varphi(a) = A$

对 $\forall g \in G, \forall x \in X, \therefore a \in X^{G \times X} \quad \therefore \varphi(a) \in (X^X)^G$

$\therefore g \in G \quad \therefore \varphi(a)(g) \in X^X \quad \therefore x \in X \quad \therefore \varphi(a)(g)(x) \in X$

$\therefore \varphi(a)(g)(x) = a(g, x) \quad \therefore a(g, x) = \varphi(a)(g)(x) \in X$

$\therefore a(g, x) = \varphi(a)(g)(x) = A(g)(x)$

$\therefore A \in (G_X)^G, g \in G \quad \therefore A(g) \in G_X \quad \therefore A(g)(x) \in X$

$\therefore A$ 是 $G \rightarrow G_X$ 的群同态 $\therefore A(1_G) = 1_{G_X} = \text{id}_X$

对 $\forall x \in X$, 有: $\therefore 1_G \in G, x \in X \quad \therefore a(1_G, x) = A(1_G)(x)$

$\therefore a(1_G, x) = A(1_G)(x) = \text{id}_X(x) = x$

对 $\forall g_1, g_2 \in G, \forall x \in X$, 有: $\therefore g_1, g_2 \in G \quad \therefore g_1 g_2 \in G$

$\therefore A$ 是 $G \rightarrow G_X$ 的群同态 $\therefore A(g_1 g_2) = A(g_1) \circ A(g_2) \in G_X$

$\therefore g_1 g_2 \in G, x \in X \quad \therefore a(g_1 g_2, x) = A(g_1 g_2)(x) = (A(g_1) \circ A(g_2))(x)$

$= A(g_1)(A(g_2)(x)) = A(g_1)(a(g_2, x)) = a(g_1, a(g_2, x))$

(注意: $a(g_2, x) \in X$)

$\therefore a(g_1, a(g_2, x)) = a(g_1 g_2, x)$

$\therefore a \in \text{Act}(G, X) \quad \therefore \varphi(a) \in \varphi(\text{Act}(G, X))$

$\therefore A \in \varphi(\text{Act}(G, X)) \quad \therefore \text{Hom}(G, G_X) \subseteq \varphi(\text{Act}(G, X))$

$$\therefore \varphi(\text{Act}(G, X)) = \text{Hom}(G, G_X)$$

定义映射 $\psi: \text{Act}(G, X) \rightarrow \text{Hom}(G, G_X)$

$$a \mapsto \varphi(a)$$

则有: 映射 $\psi: \text{Act}(G, X) \rightarrow \text{Hom}(G, G_X)$ 良定且是双射.

定理: $X^{G \times X}$ 和 $(X^X)^G$ 是两个集合,

$$\varphi: X^{G \times X} \xrightarrow{1:1} (X^X)^G$$

是一个自然的双射.

$$a \mapsto \varphi(a): G \rightarrow X^X$$

$$g \mapsto \varphi(a)(g): X \rightarrow X$$

$$x \mapsto a(g, x)$$

已知 $\text{Act}(G, X) \subseteq X^{G \times X}$, 已经证明了 $\text{Hom}(G, G_X) \subseteq (X^X)^G$,

已经证明了 $\varphi(\text{Act}(G, X)) = \text{Hom}(G, G_X)$. 定义映射 ψ 为:

$\psi: \text{Act}(G, X) \rightarrow \text{Hom}(G, G_X)$. 则有: 映射 ψ 良定且是双射.

$$a \mapsto \varphi(a)$$

Proof. 调用 20250816-() / 第三章 / 习题解答 / 3 笔记中的一个引理. $\square$

定义(群作用忠实) 群 G 作用在非空集合 X 上. 若 $\bigcap_{x \in X} \text{Stab}_G(x) = \{1_G\}$,

则称此群作用忠实.

~~Lemma: 群 G 作用在非空集合 X 上, 则有:~~

~~群作用忠实 $\Leftrightarrow$ 对 $\forall g \in G$, 都有 (对 $\forall x \in X$)~~

Lemma: 群 G 作用在非空集合 X 上, 则有:

$$\text{群作用忠实} \Leftrightarrow \forall g \in G \left((\forall x \in X, a(g, x) = x) \Rightarrow g = 1_G \right)$$

Proof: $(\Rightarrow)$: 对 $\forall g \in G$,

若对 $\forall x \in X$ 都有 $a(g, x) = x$, 则对 $\forall x \in X$ 都有 $gx = x$

$$\therefore \text{对 } \forall x \in X \text{ 都有 } g \in \text{Stab}_G(x) \quad \therefore g \in \bigcap_{x \in X} \text{Stab}_G(x) = \{1_G\}$$

$$\therefore g = 1_G \quad \therefore (\forall x \in X, a(g, x) = x) \Rightarrow g = 1_G$$

$(\Leftarrow)$: 对 $\forall x \in X$, $\therefore 1_G x = x \quad \therefore 1_G \in \text{Stab}_G(x)$

$$\therefore 1_G \in \bigcap_{x \in X} \text{Stab}_G(x) \quad \therefore \{1_G\} \subseteq \bigcap_{x \in X} \text{Stab}_G(x) \subseteq G$$

对 $\forall g \in \bigcap_{x \in X} \text{Stab}_G(x)$, 有: $\overset{g \in G, \text{且}}{\forall} \text{对 } \forall x \in X, g \in \text{Stab}_G(x)$

$$\overset{g \in G, \text{且}}{\forall} \text{对 } \forall x \in X, gx = x \quad \therefore g = 1_G \quad \therefore g \in \{1_G\}$$

$$\therefore \bigcap_{x \in X} \text{Stab}_G(x) \subseteq \{1_G\} \quad \therefore \bigcap_{x \in X} \text{Stab}_G(x) = \{1_G\}$$

$\therefore$ 群作用忠实. $\square$

Lemma: 群 G 作用在非空集合 X 上, (记这个群作用为 a , 即 $a \in \text{Act}(G, X)$)

则有: 群作用 a 忠实 $\Leftrightarrow \psi(a) = \varphi(a)$ 是 $G \rightarrow G_X$ 的单同态.

Proof: $\because a$ 是群 G 在非空集合 X 上的群作用 $\therefore a \in \text{Act}(G, X)$

$$\therefore \psi(a) = \varphi(a) \in \text{Hom}(G, G_X) \quad \therefore \psi(a) = \varphi(a) \text{ 是 } G \rightarrow G_X \text{ 的群同态.}$$

($\Rightarrow$): 对 $\forall g_1, g_2 \in G$, 若 $\varphi(\alpha)(g_1) = \varphi(\alpha)(g_2)$, 则有:

$\because \varphi(\alpha)$ 是 $G \rightarrow \mathcal{G}_X$ 的群同态, $g_1, g_2 \in G \therefore \varphi(\alpha)(g_1) \in \mathcal{G}_X$,

$\varphi(\alpha)(g_2) \in \mathcal{G}_X. \therefore \varphi(\alpha)(g_1) = \varphi(\alpha)(g_2) \in \mathcal{G}_X$

$\therefore$ 对 $\forall x \in X$, 都有 $\varphi(\alpha)(g_1)(x) = \varphi(\alpha)(g_2)(x)$

$\therefore$ 对 $\forall x \in X$, 都有 $\alpha(g_1, x) = \alpha(g_2, x)$

$\because g_1, g_2 \in G \therefore g_1^{-1}g_2 \in G.$

$\therefore$ 对 $\forall x \in X$, 都有 $\alpha(g_1^{-1}g_2, x) = \alpha(g_1^{-1}, \alpha(g_2, x))$
 $= \alpha(g_1^{-1}, \alpha(g_1, x)) = \alpha(g_1^{-1}g_1, x) = \alpha(1_G, x) = x$

$\therefore g_1^{-1}g_2 \in G$, 且对 $\forall x \in X$, 都有 $\alpha(g_1^{-1}g_2, x) = x$

$\therefore g_1^{-1}g_2 = 1_G \therefore g_1 = g_2. \therefore \varphi(\alpha)$ 是 $G \rightarrow \mathcal{G}_X$ 的单同态.

($\Leftarrow$): 对 $\forall g \in G$.

若对 $\forall x \in X$ 都有 $\alpha(g, x) = x$, 则对 $\forall x \in X, \because g \in G, \varphi(\alpha)$ 是

$G \rightarrow \mathcal{G}_X$ 的群同态 $\therefore \varphi(\alpha)(g) \in \mathcal{G}_X \because x \in X \therefore \varphi(\alpha)(g)(x) \in X$

$\therefore \varphi(\alpha)(g)(x) = \alpha(g, x) = x \therefore$ 对 $\forall x \in X, \varphi(\alpha)(g)(x) = x$

$\therefore g \in G, \varphi(\alpha)$ 是 $G \rightarrow \mathcal{G}_X$ 的群同态 $\therefore \varphi(\alpha)(g) \in \mathcal{G}_X$

$\therefore \varphi(\alpha)(g) \in \mathcal{G}_X$, 且对 $\forall x \in X, \varphi(\alpha)(g)(x) = x$

$\therefore \varphi(\alpha)(g) = id_X \because \varphi(\alpha)$ 是 $G \rightarrow \mathcal{G}_X$ 的群同态 $\therefore \varphi(\alpha)(1_G) = 1_{\mathcal{G}_X}$

$\therefore \varphi(\alpha)(1_G) = 1_{\mathcal{G}_X} = id_X \therefore \varphi(\alpha)(g) = \varphi(\alpha)(1_G)$

$\therefore \varphi(\alpha)$ 是 $G \rightarrow \mathcal{G}_X$ 的单同态 $\therefore g = 1_G \therefore$ 群作用 α 忠实 $\square$

定义 (群作用自由) 群 G 作用在非空集合 X 上. 若对 $\forall x \in X$ 都有 $\text{Stab}_G(x) = \{1_G\}$, 则称此群作用自由.

Lemma: 群 G 作用在非空集合 X 上, 则对 $\forall x \in X$, 都有 $1_G \in \text{Stab}_G(x)$.

Proof: 对 $\forall x \in X$, 有: $\text{Stab}_G(x) = \{g \in G \mid gx = x\} \subseteq G$.

$\because 1_G \in G$ 且 $1_G x = x \quad \therefore 1_G \in \text{Stab}_G(x) \quad \square$

Lemma: 群 G 作用在非空集合 X 上, 且这个群作用自由. 则有: 这个群作用忠实.

Proof: 对 $\forall x \in X$. $\because$ 这个群作用自由 $\therefore \text{Stab}_G(x) = \{1_G\}$

$\therefore$ 对 $\forall x \in X$, 都有 $\text{Stab}_G(x) = \{1_G\}$

$\therefore \bigcap_{x \in X} \text{Stab}_G(x) = \bigcap_{x \in X} \{1_G\} = \{1_G\}$

$\therefore$ 这个群作用忠实 $\square$

Lemma: 群 G 作用在非空集合 X 上 (记这个群作用为 α , 即 $\alpha \in \text{Act}(G, X)$).

则有: 群作用 α 自由 $\Leftrightarrow \forall g \in G \quad \forall x \in X \left(\alpha(g, x) = x \Rightarrow g = 1_G \right)$

Proof: $(\Rightarrow)$: 对 $\forall g \in G, \forall x \in X$.

若 $\alpha(g, x) = x$, 则 $gx = x$. $\therefore g \in G$ 且 $gx = x \quad \therefore g \in \text{Stab}_G(x)$

$$\because x \in X \text{ 且群作用自由} \quad \therefore \text{Stab}_G(x) = \{1_G\} \quad \therefore g = 1_G$$

($\Leftarrow$): 对 $\forall x \in X$.

$$\because \text{群 } G \text{ 作用在非空集合 } X \text{ 上} \quad \therefore 1_G \in \text{Stab}_G(x) \quad \therefore \{1_G\} \subseteq \text{Stab}_G(x)$$

$$\text{对 } \forall g \in \text{Stab}_G(x), \text{ 有: } g \in G \text{ 且 } gx = x \quad \therefore g \in G \text{ 且 } a(g, x) = x.$$

$$\because g \in G, x \in X, a(g, x) = x \quad \therefore g = 1_G \quad \therefore g \in \{1_G\}$$

$$\therefore \text{Stab}_G(x) \subseteq \{1_G\} \quad \therefore \text{Stab}_G(x) = \{1_G\}. \quad \therefore \text{这个群作用自由.} \quad \square$$

Lemma (群在自身上的左平移作用) G 是一个任意的群, $\therefore G \neq \emptyset$.

G 也是一个非空集合. 定义映射 $a_L: G \times G \rightarrow G$. 则有:

$$(g, x) \mapsto gx$$

a_L 是群 G 在非空集合 G 上的一个群作用.

$$\text{Proof: 对 } \forall x \in G, \text{ 有: } a_L(1_G, x) = 1_G \cdot x = x$$

$$\text{对 } \forall g_1, g_2 \in G, \forall x \in G, \text{ 有: } a_L(g_1, a_L(g_2, x)) = a_L(g_1, g_2 x)$$

$$= g_1(g_2 x) = (g_1 g_2) x = a_L(g_1 g_2, x)$$

↓
群 G 中的乘法结合律

$\therefore a_L$ 是群 G 在 G 上的作用. $\square$

Lemma: 群 G 在自身上的左平移作用是自由的, 也是忠实的

Proof: 对 $\forall g \in G, \forall x \in G$,

$$\text{若 } a_L(g, x) = x, \text{ 则 } gx = x \quad \therefore (gx)x^{-1} = xx^{-1} = 1_G \quad \therefore g = 1_G$$

$\therefore$ 群作用 α_L 自由.

$\therefore$ 群作用 α_L 忠实.

□

定理 (Cayley's theorem, Cayley 定理) 任何群 G 都能嵌入为 S_G 的子群

Proof: $\therefore G$ 是群 $\therefore G$ 是非空集合 $\therefore S_G = \{\text{双射 } \sigma: G \rightarrow G\}$.

已经证明了 $(S_G, \circ)$ 是群.

$\therefore$ 已经证明了 $\alpha_L: G \times G \rightarrow G$ 是群 G 在非空集合 G 上的一个
 $(g, x) \mapsto gx$

群作用 $\therefore \alpha_L \in G^{G \times G}$ 且 $\alpha_L \in \text{Act}(G, G)$

$\therefore \psi(\alpha_L) = \varphi(\alpha_L) \in \text{Hom}(G, S_G)$

$\therefore \varphi(\alpha_L)$ 是 $G \rightarrow S_G$ 的群同态.

$\therefore$ 群作用 α_L 忠实 $\therefore \varphi(\alpha_L)$ 是 $G \rightarrow S_G$ 的单同态.

记 $A_L = \varphi(\alpha_L) = \psi(\alpha_L)$ $\therefore A_L$ 是 $G \rightarrow S_G$ 的单同态

$\therefore G \leq G \therefore A_L(G) \leq S_G$

$\therefore A_L: G \rightarrow A_L(G)$ 是群同构 $\therefore G \simeq A_L(G) \leq S_G$.

~~A_L~~ $\therefore A_L$ 是 $G \rightarrow S_G$ 的单同态

$\therefore A_L(G) = \{A_L(g) \mid g \in G\} \leq S_G$

$\therefore G^{G \times G} = \{\text{映射 } f: G \times G \rightarrow G\}$, $(G^G)^G = \{\text{映射 } f: G \rightarrow G^G\}$.

$\text{Act}(G, G) = \{\alpha \in G^{G \times G} \mid \alpha \text{ 是群 } G \text{ 在非空集合 } G \text{ 上的群作用}\}$.

$\text{Act}(G, G) \subseteq G^{G \times G}$

$$\mathcal{S}_G = \{\text{双射 } \sigma: G \rightarrow G\}, \quad G^G = \{\text{映射 } f: G \rightarrow G\} \quad \therefore \mathcal{S}_G \subseteq G^G$$

已经证明了 $(\mathcal{S}_G, \circ)$ 是群.

$$\text{Hom}(G, \mathcal{S}_G) = \{A \in (\mathcal{S}_G)^G \mid A \text{ 是 } G \rightarrow \mathcal{S}_G \text{ 的群同态}\}.$$

$$\therefore \text{Hom}(G, \mathcal{S}_G) \subseteq (\mathcal{S}_G)^G, \quad \text{Hom}(G, \mathcal{S}_G) \subseteq (G^G)^G$$

已经证明了存在自然的双射 $\varphi: G^{G \times G} \xrightarrow{1:1} (G^G)^G$ 为

$$\varphi: G^{G \times G} \xrightarrow{1:1} (G^G)^G$$

$$a \longmapsto \varphi(a): G \rightarrow G^G$$

$$g \mapsto \varphi(a)(g): G \rightarrow G$$

$$x \mapsto a(g, x)$$

定义映射 ψ 为 $\psi: \text{Act}(G, G) \rightarrow \text{Hom}(G, \mathcal{S}_G)$. 则映射 ψ 良定且

$$a \longmapsto \varphi(a)$$

是双射.

$$\therefore a_L \in G^{G \times G} \text{ 且 } a_L \in \text{Act}(G, G) \quad \therefore \psi(a_L) = \varphi(a_L) \in \text{Hom}(G, \mathcal{S}_G)$$

$$\therefore A_L = \varphi(a_L) = \psi(a_L) \in \text{Hom}(G, \mathcal{S}_G). \text{ 已经证明了 } A_L \text{ 是 } G \rightarrow \mathcal{S}_G \text{ 的}$$

单同态.

$$\text{对 } \forall g \in G, \quad A_L(g) = \varphi(a_L)(g) \in \mathcal{S}_G$$

$$\text{对 } \forall x \in G, \text{ 有: } A_L(g)(x) = \varphi(a_L)(g)(x) = a_L(g, x) = gx \quad \text{因此 } A_L(g) = \lambda_g$$

$$\therefore A_L(G) = \{A_L(g) \mid g \in G\} = \{\sigma \in \mathcal{S}_G \mid \exists g \in G, \text{ s.t. } \sigma = A_L(g)\}$$

$$= \{\sigma \in \mathcal{S}_G \mid \exists g \in G, \text{ s.t. 对 } \forall x \in G, \text{ 都有 } \sigma(x) = A_L(g)(x)\}$$

$$= \{\sigma \in \mathcal{S}_G \mid \exists g \in G, \text{ s.t. 对 } \forall x \in G, \text{ 都有 } \sigma(x) = gx\}$$

$$\therefore A_L : G \longrightarrow G_a$$

$$g \mapsto A_L(g) = \varphi(a_L)(g) : G \longrightarrow G$$

$$x \mapsto gx$$



Lemma: G 是群, 对 $\forall g \in G$, 定义映射 $\lambda_g : G \longrightarrow G$. 则有:
 $x \mapsto gx$.

① 对 $\forall g \in G$, 映射 λ_g 是 $G \rightarrow G$ 的 ~~群同构~~ 双射, 即 $\lambda_g \in G_a$

$$\textcircled{2} \lambda_{1_G} = id_G$$

$$\textcircled{3} \lambda_g \text{ 是 } G \rightarrow G \text{ 的群同构} \Leftrightarrow g = 1_G \Leftrightarrow \lambda_g = id_G$$

$$\textcircled{4} \text{ 对 } \forall g \in G, \text{ 有: } \lambda_g \circ \lambda_{g^{-1}} = id_G = \lambda_{g^{-1}} \circ \lambda_g$$

Proof: ①. 对 $\forall g \in G$, 映射 λ_g 的定义为 $\lambda_g : G \longrightarrow G$
 $x \mapsto gx$

$$\text{对 } \forall x \in G, \because g \in G \text{ 且 } x \in G \therefore gx \in G \therefore \lambda_g(x) = gx \in G.$$

$$\therefore \lambda_g(G) \subseteq G$$

$$\text{对 } \forall x, y \in G, \text{ 若 } x = y, \text{ 则 } gx = gy \therefore \lambda_g(x) = gx = gy = \lambda_g(y)$$

$\therefore \lambda_g : G \rightarrow G$ 是一个映射.

$$\text{对 } \forall x, y \in G, \text{ 若 } \lambda_g(x) = \lambda_g(y), \text{ 则 } gx = gy \because g \in G \therefore g^{-1} \in G$$

$$\therefore g^{-1}(gx) = g^{-1}(gy) \therefore x = y \therefore \lambda_g : G \rightarrow G \text{ 是一个单射.}$$

$$\text{对 } \forall y \in G, \because g \in G \therefore g^{-1}y \in G. \therefore \lambda_g(g^{-1}y) = g(g^{-1}y) = (gg^{-1})y$$

$$= 1_G y = y \therefore \lambda_g : G \rightarrow G \text{ 是一个满射} \therefore \lambda_g : G \rightarrow G \text{ 是一个双射.}$$

$$\because \lambda_g \in \mathcal{E}_G \quad \because \{\lambda_g \mid g \in G\} \subseteq \mathcal{E}_G. \quad \textcircled{1} \text{得证.}$$

$$\textcircled{2}. \because 1_G \in G \quad \because \lambda_{1_G} : G \rightarrow G$$

$$x \mapsto 1_G \cdot x = x$$

$$\text{又} \forall x \in G, \lambda_{1_G}(x) = 1_G \cdot x = x = \text{id}_G(x) \quad \therefore \lambda_{1_G} = \text{id}_G. \quad \textcircled{2} \text{得证}$$

$$\textcircled{3}: \text{若 } g = 1_G, \text{ 则 } \lambda_g = \lambda_{1_G} = \text{id}_G.$$

$$\text{若 } \lambda_g = \text{id}_G, \text{ 则 } \because 1_G \in G \quad \therefore 1_G = \text{id}_G(1_G) = \lambda_g(1_G) = g \cdot 1_G = g$$

$$\therefore g = 1_G \quad \therefore \text{右边的} \Leftarrow \text{得证.}$$

$$\text{若 } g = 1_G, \text{ 则 } \lambda_g = \lambda_{1_G} = \text{id}_G \text{ 是 } G \rightarrow G \text{ 的群同构. 左边的} \Leftarrow \text{得证.}$$

$$\text{若 } \lambda_g \text{ 是 } G \rightarrow G \text{ 的群同构, 则 } \lambda_g \text{ 是 } G \rightarrow G \text{ 的群同态}$$

$$\therefore \lambda_g(1_G) = \lambda_g(1_G \cdot 1_G) = \lambda_g(1_G) \cdot \lambda_g(1_G)$$

$$\therefore g \cdot 1_G = (g \cdot 1_G) \cdot (g \cdot 1_G) \quad \therefore g = g^2 \quad \because g \in G \quad \therefore g^{-1} \in G$$

$$\therefore g^{-1} \cdot g = g^{-1} \cdot g^2 \quad \therefore 1_G = g \quad \therefore g = 1_G. \quad \text{左边的} \Rightarrow \text{得证.}$$

$$\textcircled{4}. \text{又} \forall g \in G. \quad \because g \in G \quad \therefore g^{-1} \in G$$

$$\therefore \lambda_g : G \rightarrow G \quad \lambda_{g^{-1}} : G \rightarrow G$$

$$x \mapsto gx \quad x \mapsto g^{-1}x$$

$$\text{已经证明了 } \lambda_g \text{ 是 } G \rightarrow G \text{ 的双射, } \lambda_{g^{-1}} \text{ 也是 } G \rightarrow G \text{ 的双射.}$$

$$\therefore \lambda_g \circ \lambda_{g^{-1}} \text{ 是 } G \rightarrow G \text{ 的双射. } \lambda_{g^{-1}} \circ \lambda_g \text{ 是 } G \rightarrow G \text{ 的双射.}$$

$$\text{又} \forall x \in G, \therefore (\lambda_g \circ \lambda_{g^{-1}})(x) = \lambda_g(\lambda_{g^{-1}}(x)) = \lambda_g(g^{-1}x) = g(g^{-1}x)$$

$$= (g g^{-1}) x = 1_G x = x = \text{id}_G(x) \quad \therefore \lambda_g \circ \lambda_{g^{-1}} = \text{id}_G$$

$$\text{对 } \forall x \in G, \quad \therefore (\lambda_{g^{-1}} \circ \lambda_g)(x) = \lambda_{g^{-1}}(\lambda_g(x)) = \lambda_{g^{-1}}(gx) = g^{-1}(gx)$$

$$= (g^{-1}g)x = 1_G x = x = \text{id}_G(x) \quad \therefore \lambda_{g^{-1}} \circ \lambda_g = \text{id}_G$$

$$\therefore \lambda_g \circ \lambda_{g^{-1}} = \text{id}_G = \lambda_{g^{-1}} \circ \lambda_g \quad \textcircled{4} \text{得证.}$$

Lemma: G 是群, 对 $\forall g \in G$, 定义映射 $\lambda_g: G \rightarrow G$. 则有:

$$x \mapsto gx$$

$$\text{对 } \forall g_1, g_2 \in G, \text{ 有: } \lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2}$$

Proof: 对 $\forall g_1, g_2 \in G$. 有: $g_1 g_2 \in G$.

$$\begin{array}{lll} \therefore \lambda_{g_1}: G \rightarrow G & \lambda_{g_2}: G \rightarrow G & \lambda_{g_1 g_2}: G \rightarrow G \\ x \mapsto g_1 x & x \mapsto g_2 x & x \mapsto (g_1 g_2) x \end{array}$$

已经证明了 $\lambda_{g_1}, \lambda_{g_2}, \lambda_{g_1 g_2}$ 都是 $G \rightarrow G$ 的双射.

$\therefore \lambda_{g_1} \circ \lambda_{g_2}$ 是 $G \rightarrow G$ 的双射.

$$\begin{aligned} \text{对 } \forall x \in G, \text{ 有: } (\lambda_{g_1} \circ \lambda_{g_2})(x) &= \lambda_{g_1}(\lambda_{g_2}(x)) = \lambda_{g_1}(g_2 x) \\ &= g_1(g_2 x) = (g_1 g_2)x = \lambda_{g_1 g_2}(x) \quad \therefore \lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2} \quad \square \end{aligned}$$

Lemma: G 是群, 对 $\forall g \in G$, 定义映射 $\lambda_g: G \rightarrow G$. 定义集合

$$x \mapsto gx$$

$$\lambda(G) = \{\lambda_g \mid g \in G\}. \text{ 则有: } \lambda(G) \leq \mathcal{S}_G.$$

Proof: 任取 $\lambda(G)$ 中的一元: λ_g (其中 $g \in G$). 有: λ_g 是 $G \rightarrow G$ 的映射, 定义为 $\lambda_g: G \rightarrow G$ $x \mapsto gx$ $\because$ 已经证明了映射 λ_g 是 $G \rightarrow G$ 的双射

$\therefore \lambda_g \in \mathcal{S}_G \quad \therefore \lambda(G) \subseteq \mathcal{S}_G$. 已经证明了 $(\mathcal{S}_G, \circ)$ 是群.

$\therefore |_{\mathcal{S}_G} = id_G = \lambda_{1_G} \in \lambda(G)$

任取 $\lambda(G)$ 中的两个元素 $\lambda_{g_1}, \lambda_{g_2}$ (其中 $g_1, g_2 \in G$)

$\because g_1, g_2 \in G \quad \therefore \lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2} \in \lambda(G)$

$\therefore \lambda(G)$ 对映射的合成封闭.

任取 $\lambda(G)$ 中的一元: λ_g (其中 $g \in G$). $\because g \in G \quad \therefore g^{-1} \in G$

$\therefore \lambda_g \in \mathcal{S}_G, \lambda_{g^{-1}} \in \mathcal{S}_G. \quad \therefore \lambda_g \circ \lambda_{g^{-1}} = id_G = \lambda_{g^{-1}} \circ \lambda_g$

$\therefore \lambda_g \circ \lambda_{g^{-1}} = |_{\mathcal{S}_G} = \lambda_{g^{-1}} \circ \lambda_g \quad \therefore (\lambda_g)^{-1} = \lambda_{g^{-1}}$

$\therefore (\lambda_g)^{-1} = \lambda_{g^{-1}} \in \lambda(G) \quad \therefore \lambda(G)$ 对取逆封闭.

$\therefore \lambda(G) \leq \mathcal{S}_G. \quad \square$

定理 (Cayley's theorem, Cayley 定理) G 是群, 对 $\forall g \in G$, 定义映射

$\lambda_g: G \rightarrow G$ 定义集合 $\lambda(G) = \{\lambda_g \mid g \in G\}$. 则已经证明了 $x \mapsto gx$

$\lambda(G) \leq \mathcal{S}_G$. 定义映射 $T: G \rightarrow \lambda(G)$ 则有: T 是 $G \rightarrow \lambda(G)$ 的群同构.
 $g \mapsto \lambda_g$

Proof: 对 $\forall z \in G$, $\because z \in G \quad \therefore \lambda_z \in \lambda(G)$

$$\therefore T(z) = \lambda_z \in \lambda(G) \quad \therefore T(G) \subseteq \lambda(G)$$

对 $\forall z_1, z_2 \in G$,

$$\text{若 } z_1 = z_2, \text{ 则 } \because z_1 = z_2 \in G \quad \therefore \lambda_{z_1}: G \rightarrow G, \lambda_{z_2}: G \rightarrow G$$
$$x \mapsto z_1 \cdot x \quad \quad \quad x \mapsto z_2 \cdot x$$

$$\text{对 } \forall x \in G, \text{ 有: } \lambda_{z_1}(x) = z_1 \cdot x = z_2 \cdot x = \lambda_{z_2}(x) \quad \therefore \lambda_{z_1} = \lambda_{z_2}$$

$$\therefore T(z_1) = \lambda_{z_1} = \lambda_{z_2} = T(z_2) \quad \therefore T: G \rightarrow \lambda(G) \text{ 是一个映射.}$$

$$\text{若 } T(z_1) = T(z_2), \text{ 则 } \lambda_{z_1} = \lambda_{z_2} \quad \therefore \lambda_{z_1} = \lambda_{z_2} \in \mathcal{S}_G.$$

$$\therefore \text{对于 } 1_G \in G, \text{ 有: } \lambda_{z_1}(1_G) = \lambda_{z_2}(1_G) \quad \therefore z_1 \cdot 1_G = z_2 \cdot 1_G$$

$$\therefore z_1 = z_2 \quad \therefore T: G \rightarrow \lambda(G) \text{ 是一个单射.}$$

$$\text{任取 } \lambda(G) \text{ 中的一元: } \lambda_g \text{ (其中 } g \in G) \quad \because g \in G \quad \therefore T(g) = \lambda_g$$

$$\therefore T: G \rightarrow \lambda(G) \text{ 是一个满射.} \quad \therefore T: G \rightarrow \lambda(G) \text{ 是一个双射.}$$

$$\text{对 } \forall z_1, z_2 \in G, \text{ 有: } z_1 \cdot z_2 \in G \quad \therefore z_1 z_2 \in G$$

$$\therefore T(z_1 \cdot z_2) = T(z_1 z_2) = \lambda_{z_1 z_2} = \lambda_{z_1} \circ \lambda_{z_2} = T(z_1) \circ T(z_2)$$

$$\therefore T: G \rightarrow \lambda(G) \text{ 是一个群同态} \quad \therefore T: G \rightarrow \lambda(G) \text{ 是一个群同构.}$$

$$\therefore T: G \cong \lambda(G) \quad \therefore G \cong \lambda(G) \leq \mathcal{S}_G \quad \square$$

Remark: 之前证明了 $G \cong A_L(G) \leq \mathcal{S}_G$.

$$A_L(G) = \{ \sigma \in \mathcal{S}_G \mid \exists g \in G, \text{ s.t. 对 } \forall x \in G, \text{ 都有 } \sigma(x) = gx \}$$
$$= \{ \sigma \in \mathcal{S}_G \mid \exists g \in G, \text{ s.t. 对 } \forall x \in G, \text{ 都有 } \sigma(x) = \lambda_g(x) \}$$

$$= \{ \sigma \in \mathcal{S}_G \mid \exists g \in G, \text{ s.t. } \sigma = \lambda_g \} = \{ \lambda_g \mid g \in G \} = \lambda(G)$$

$$\therefore A_L(G) = \lambda(G)$$

之前写出的单同态 $A_L : G \rightarrow \mathcal{S}_G$

$$g \mapsto A_L(g) = \varphi(\alpha_L)(g) : G \rightarrow G$$

$$x \mapsto gx$$

$$\therefore A_L : G \rightarrow \mathcal{S}_G$$

$$g \mapsto \lambda_g$$

$$\therefore A_L = T \text{ (忽略陪域的差异)}$$